

Write legibly showing all the steps VERY clearly. (Total points = 100).

- (1) (a) **State** precisely the *Dominated convergence Theorem (DCT)* (Not the EDCT!).
 (b) **Prove** that the conclusion of DCT remains true even if “ $f_n \rightarrow f$ a.e(μ)” is replaced by “ $f_n \rightarrow f$ in measure” in the statement.
 (c) Give an example of a function f that is **not** $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.
 (d) Give an example of a function f that is **not** $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable, but $|f|$ is $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

(5+10+10+10=35 points)

- (2) (a) Argue that there exists a *random variable* (call it ϵ) on some probability space $(\Omega, \mathcal{F}, P)$ whose distribution on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is Normal(0,1) measure. [Note: $\int \epsilon dP = 0$ in this case - you do not need to prove that].
 (b) Let for $i = 1, \dots, n$, ϵ_i be a *random variables* defined on some probability space $(\Omega, \mathcal{F}, P)$ whose distribution is Normal(0,1). Let $\{x_i\}_{i \geq 1}$ be a sequence of real numbers (with $\sum x_i^2 > 0$). For some $\beta \in \mathbb{R}$, define functions on $(\Omega, \mathcal{F}, P)$:

$$Y_i(\omega) = x_i\beta + \epsilon_i(\omega), \quad \omega \in \Omega, \quad i = 1, \dots, n.$$

and

$$\hat{\beta}_n(\omega) = \frac{\sum_{i=1}^n x_i Y_i(\omega)}{\sum_{i=1}^n x_i^2}, \quad \omega \in \Omega.$$

Argue Y_i and $\hat{\beta}_n$ is real valued *measurable* function on $(\Omega, \mathcal{F}, P)$ (i.e. random variables).

- (c) Compute $\int Y_i dP$ and $\int \hat{\beta}_n dP$ clearly stating properties of integrations used.

(5+10+10=25 points)

- 3 Let m be the Lebesgue measure on $[0, 1]$ and $I_A(x)$ denote the indicator function of A . Consider the following sequence of random variables on $(\Omega = [0, 1], \mathcal{F} = \mathcal{B}[0, 1], P = m)$: For $n \geq 1$, $X_n(\omega) = n^2 I_{[\frac{n-1}{n}, 1]}(\omega)$ and $X(\omega) = 0$ for $\omega \in \Omega$.

- (i) Show $X_n \rightarrow X$ almost surely (P).
 (ii) Does $X_n \rightarrow X$ in probability ?
 (iii) Show $X_n \rightarrow X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < \frac{1}{2}$, but this convergence does not hold for $\frac{1}{2} \leq p < \infty$.
 (iv) Show that, in general, $X_n \rightarrow X$ in L^p for some $0 < p < \infty$ implies that $X_n \rightarrow X$ in L^q , for any $0 < q < p$.

(10+10+10+10 = 40 points)